

SAIFELDIN MOHAMED SAIED

Software Engineering Student

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EDUCATION

B.Sc. Software Engineering — Egyptian Chinese University (ECU) 2025 – Present · Cairo
Previously: Cairo University, Faculty of Computers & Artificial Intelligence (2024 – 2025)
American Diploma — Dome International Schools Graduated 2024 · Giza

EXPERIENCE

Generative AI Foundation Program — Commercial International Bank (CIB), Egypt Aug 2026 · Cairo

- Completed an intensive, month-long training program on Generative AI, LLMs, foundation models, and prompt engineering as part of the CIB Summer Internship 2026.
- Explored strategic GenAI applications in financial services — customer experience, digital banking transformation, and fraud detection.

IT Summer Trainee — Konooz Misr for Marble and Granite Jun – Aug 2024 · Cairo

- Networking tasks: cable management, switch configuration, and connectivity troubleshooting.
- Contributed to internal software development projects and collaborated with the IT team.

PROJECTS

3D Maze Game — Graphics Programming C++ · OpenGL · FreeGLUT

- Hand-coded 4x4 matrix math** (translation, Rodrigues' axis rotation, scaling) and homogeneous coordinates — **no external math libraries**.
- Blinn-Phong illumination** with 2 light sources, perspective & orthographic projections, and 3 switchable camera modes via a **hierarchical matrix-stack** model system.

Energy Consumption Prediction & Analysis System — Machine Learning Python · CatBoost · PyTorch · Scikit-learn

- Engineered features through IQR-based outlier removal, time-based and cyclical extraction, and StandardScaler normalization for robust model input.
- CatBoost Regressor** trained and benchmarked against a custom **PyTorch neural network**, evaluated using MAE, RMSE, and R^2 .
- K-Means Clustering** with Elbow Method and Silhouette Score applied to uncover distinct household consumption patterns.

Ei Gehad Automotive — Full-Stack Dealership Platform Node.js · Express · SQLite · JWT

- 71-vehicle catalog** with client-side search/filter, paginated "load more," per-vehicle detail pages, and **JWT-secured admin CRUD**.
- Custom Wikimedia Commons API scraper** with a token/year-proximity scoring heuristic to correctly match real photos for all 71 models.
- Zero-backend GitHub Pages deployment** via a bundled JSON fallback; validated with **34 passing end-to-end tests**.

Warehouse Multi-Robot Coordination — AI Course Python · A* · MAPF

- A* pathfinding** for autonomous robots on 10x10 grid; Cooperative MAPF with Prioritized Planning resolves vertex/edge conflicts.
- Compared Independent A* vs MAPF across makespan, collisions, and computation time; real-time matplotlib visualisation.

Study Planner — Full-Stack Web Application Python · Flask · SQLite · Bootstrap · JS

- Flask backend with **SHA-256 auth**, session management, REST API; SQLite DB with 3 tables and foreign key constraints.
- Priority scheduling algorithm** (High/Medium/Low + earliest deadline) auto-assigns study blocks with 15-min breaks — reduces scheduling from 30+ min to zero.

CERTIFICATIONS

- NovaStar Certified LED Screen Technician (NCE)** — NovaStar Technologies May 2026 · Valid 2 yrs · No. 2026IZZ1066
- AI Foundations Knowledge Badge** — SAS 2026 · Credly Verified
- Ethical Hacking** Aug 2024 · CcZSI2nM5
- Computer Network Fundamentals** Aug 2024 · 9htP2j7L6i
- Introduction to Network Security** Sep 2024 · JB2VGzNPmY
- Python Programming Basics** Sep 2024 · EkZIKEQfEe
- Implementation of Computer Network Fundamentals** Sep 2024 · ZCfw8IHmFM

Mahara-Tech (Ministry of Communications, Egypt) — last 5 certificates above

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, HTML5, CSS3
OOP: Inheritance, Polymorphism, Encapsulation, Abstract Classes, Virtual Functions, Structs
AI/Algo: A* Pathfinding, MAPF, Prioritized Planning, Machine Learning, Deep Learning, K-Means Clustering
Web Dev: Flask, Node.js, Express, REST API, SQLite, JWT, Bootstrap 5, Chart.js
Other: OpenGL, GLSL, Networking, Cybersecurity, Ethical Hacking, Software Testing

LANGUAGES

German (Intermediate) · English (Proficient) · Arabic (Native)